Devontae C. Baxter

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Research Identity

The Impact of Environment on Galaxy Evolution:
– Mapping the 3D cosmic web to correlate galaxy properties with their local and large-scale environments;
– Constraining the relative roles of internal feedback and external processes in regulating galaxy growth and quenching;
– Tracing the emergence of environmental influences on galaxy evolution back to the earliest protoclusters.
Collaborations:
– Charting Cluster Construction with VUDS and ORELSE (C3VO);
– Gemini Observations of Galaxies in Rich Early ENvironments (GOGREEN);
– The Grism Reionization And Cosmic Evolution Survey (GRACE);
Unified Astronomy Thesaurus Keywords: Galaxy evolution, Galaxy quenching, High-redshift galaxy clusters.

Professional Appointments

Future Faculty in the Physical Sciences Fellow at Princeton University — Aug 2026 - Present
Faculty Mentor: Jenny E. Greene

NSF Astronomy & Astrophysics Postdoctoral Fellow at UC San Diego — July 2023 - July 2026
Faculty Mentor: Alison L. Coil

Education

The University of California, Irvine — Sep 2017 - June 2023
Ph.D. in Physics | Thesis Advisor: Michael C. Cooper

Georgia Institute of Technology — Jan 2015 - May 2017
B.S. in Physics with highest honors | Research Advisor: Alberto Fernandez-Nieves

Grants, Fellowships & Awards

Summary: Total funding awarded (accepted and declined) **$2.2M**

Grants

Roman Space Telescope Cycle 1 Program ($1.2M, Program ID# 2003, PI: S. Malhotra, **co-PI: D. C. Baxter**) — 2026
GRACE: The Grism Reionization And Cosmic Evolution Survey

National Science Foundation ($330k, Award# 2303800, **PI: D. C. Baxter**) — 2023
Constraining the Cosmic Evolution of Environmental Quenching & Predicting Protocluster Populations

Prize Fellowships

Princeton Future Faculty in the Physical Sciences Fellowship ($302k) — 2026

UC Chancellor's Postdoctoral Fellowship ($10k) — 2023

Sullivan Prize Postdoctoral Fellowship ($240k, Declined) — 2023

Other Honors & Awards

UC San Diego Bouchet Honor Society ($500)	2026
Honorable Mention: UC Irvine President's Dissertation Year Fellowship ($1k)	2022
AAS National Osterbrock Leadership Program Fellowship	2020
UCI Physics & Astronomy Graduate Student Mentor Award	2020
LSSTC Data Science Fellowship	2019
Honorable Mention: Ford Foundation Predoctoral Fellowship	2019
UCI Eugene Cota-Robles Graduate Fellowship ($67k)	2017
Tulsa Community Foundation: AMC Cares Scholarship ($5k)	2015
Georgia Hope Scholarship ($15k)	2015

Publication Statistics (full list attached at the end)

Peer-reviewed papers: **18** (**6** first-author, **2** second-author, **10** contributing author) | Citations: **240+**; h-index=**9**

Teaching Experience

Summary: *Taught* ***6*** *courses (**1** Instructor of Record,* ***3*** *Teaching Assistant,* ***2*** *Guest Lecturer) and* ***12*** *workshops.*

Instructor of Record

ASTR 20A (Introduction to Astrophysics I) at UC San Diego — Fall 2025

Teaching Assistant

Physics 20E (Life in the Universe) at UC Irvine — Spring 2019
Physics 20B (Cosmology) at UC Irvine — Winter 2019
Physics 20D (Space Science) at UC Irvine — Fall 2018

Guest Lecturer

ASTR 2 (Galaxies and the Universe) at UC San Diego — May 2025
Physics 138 (Astrophysics of Galaxies) at UC Irvine — Nov 2020

Workshops & Bootcamps

STARTastro Python Workshop at UC San Diego — 2024-2026
Computational Astrophysics Research Preparation (CARP) Python Workshop at UC San Diego — 2024-2026
Computational Research Access NEtwork (CRANE) Astronomy Data Analysis Workshop — 2023-2025
UC Carpentries Fall Workshop Series at UC San Diego — Sep 2023
LSSTC Data Science Fellowship Program Workshop at University of Washington — June 2023
Physics Olympiad Bootcamp at Ardent Academy — Dec 2019

Training & Certificates

Instructor Certification from *The Carpentries* — 2024

The Carpentries is an international organization that teaches data science skills through hands-on workshops. This certificate **recognizes the completion of pedagogical training to teach these skills to a global population of learners.**

Certification in Course Design — 2020

This certificate from UCI Division of Teaching Excellence and Innovation **recognizes the completion of training to design effective, inclusive, and engaging courses.**

Certification for the Integration of Research, Teaching and Learning — 2020

This certificate from UCI Division of Teaching Excellence and Innovation **recognizes the completion of training to integrate evidence-based teaching practices to enhance student outcomes.**

Mentoring Excellence Program Certification — 2019

This certificate from UCI Graduate Division **demonstrates acquisition of mentoring skills, such as cultural competence and inclusivity, effective communication, navigating power dynamics, and conflict mediation.**

Professional Talks

Summary: ***33*** *presentations (**19** invited,* ***14*** *contributed) | †=invited; ∗=contributed*

Colloquia/Seminars

University of Washington Colloquium† — April 2026
University of Southern California CosmoLab Seminar† — Mar 2026
Cal State Long Beach Colloquium† — Mar 2026
UC San Diego Community and Science Advancements in Spanish (CaSAS) Seminar† — Sep 2025
Caltech Tea Talk† — May 2025
University of Wisconsin-Madison Science Seminar† — April 2025
UC Irvine Astronomy Seminar† — April 2025
UC Riverside Astronomy Seminar† — April 2025
UC San Diego Simulations TheoRy AND more (STRAND) Meeting† — March 2025
Washington State University Physics & Astronomy Colloquium† [video] — Nov 2024
University of Florida Astronomy Colloquium† — Nov 2024
UC Davis Cosmology and Astronomy Seminar† — Oct 2024
Georgia Institute of Technology Center for Relativistic Astrophysics Seminar† — Nov 2023

UC San Diego Astronomy Colloquium† Oct 2023
University of Kansas Astronomy Seminar† April 2022
The Ohio State University Little Galaxies Journal Club† March 2021

Conferences/Workshops

GalScale2026* | Ascona, Switzerland July 2026
NSF AAPF Fellows Symposium* | Phoenix, AZ Jan 2026
Galaxy Formation & Evolution in Southern California 2025* | San Diego, CA Sep 2025
The Baryon Cycle in the Cosmic Web* | Bern, Switzerland Sep 2025
Tracing Cosmic Evolution with Galaxy Clusters V* | Sesto, Italy July 2025
President's Postdoctoral Fellowship Program Academic Spring Retreat* | Lake Arrowhead, CA April 2025
Conference for Emerging Black Academics in STEM* | Pasadena, CA April 2025
Ensenada San Diego Astrophysical Society Meeting* | Ensenada, Mexico Sep 2024
Galaxy Formation & Evolution in Southern California 2024* | Pasadena, CA Sep 2024
President's Postdoctoral Fellowship Program Academic Spring Retreat* | Lake Arrowhead, CA April 2024
AAS 243rd Meeting (Dissertation Talk)* | New Orleans, LA Jan 2024
NSF AAPF Fellows Symposium* | New Orleans, LA Jan 2024
The Baryon Cycle of Protocluster Galaxies† | Leiden, Netherlands Oct 2023
First Structures in the Universe 2023† | Paris, France Sept 2023
Inaugural Conference for Emerging Black Academics in STEM† | Pasadena, CA May 2023
AAS 240th Meeting* | Pasadena, CA June 2022
Georgia Tech College of Science REU Poster Symposium* | Atlanta, GA July 2017

Advising & Mentorship

Summary: *Supervised **2** undergraduate students; mentored **21** students (**4** graduate, **17** undergraduate).*

Students Supervised

Sophia Um (UC San Diego, supervised senior thesis; now graduate student at San Diego State University)
Maxwell Beckman (UC San Diego, supervised undergraduate research project)

Academic Mentorship

UC San Diego Astronomy & Astrophysics Postdoc × Grad Mentorship Program | **2 grads**
Computational Astrophysics Research Preparation Program | **17 undergrads**
UC Irvine PACE Mentorship Program Peer Mentor | **2 grads**

Service & Leadership

Summary: *Organized **3** conferences/workshops; Served on **2** departmental/student committees; Reviewer for **3** grant/telescope proposals; Developed and/or led **3** mentoring/outreach programs.*

Scientific Organizing

Galaxy Formation & Evolution in Southern California Conference 2025
23rd Annual NSF AAPF Postdoc Symposium 2024
"Cultivating the Next Generation of Leaders" Splinter Session at the 240th AAS Meeting 2022

Departmental Committees

UCSD Astronomy & Astrophysics Colloquium Committee Member 2024-2026

Grant & Telescope Proposal Review

Hubble Space Telescope Cycle 33 TAC Expert Reviewer 2025
Future Investigators in NASA Earth and Space Science and Technology (FINESST) No-Panel Reviewer 2025
Hubble Space Telescope Cycle 32 TAC Expert Reviewer 2024

Academic Leadership

Founder and Program Director - Computational Astrophysics Research Preparation (CARP) 2023-Present
NSF-supported coding and mentorship program for community college transfer students.

Program Coordinator - Computational Research Access NEtwork (CRANE) 2022-2025
International coding and mentorship program for historically underrepresented scholars.

Co-Chair - UCI Physics & Astronomy Community Excellence (PACE) Peer Mentoring Program 2020-2023
Student-led program focused on 1-on-1 and group mentoring for first-year graduate students.

Social Event Chair - UC Irvine Physics Graduate Caucus 2018-2020

Outreach

Invited Speaker - Cosmos & Coffee Astronomy Talk Series May 2026
Instructor - AstroReach Astronomy Essentials Course at The Preuss School Jan 2026
Invited Speaker - Peninsula Astronomical Society Talk Series Jan 2026
Invited Speaker - Inclusive Pedagogy Session at Choir Conference July 2025
Invited Speaker - Lick Observatory Summer Series Public Outreach Talks [video] July 2025
Invited Speaker - AstroReach San Diego Feb 2025
Volunteer - UC San Diego Cosmic Tours Portable Planetarium Shows 2025-2026
Invited Guest - Astro[sound]bites Podcast Episode on Machine Learning in Astrophysics Oct 2020
Volunteer - UC Irvine Telescope Outreach Program 2019-2020

Observing Experience

Summary: **17.5 nights** on Keck (KCWI, KCRM, MOSFIRE, DEIMOS) as Co-Investigator across **5** programs.

Observatory / Instrument	Time	Program	Role
Keck/KCWI + KCRM	1.5 night	Illuminating the Physics of the Multiphase Baryon Cycle	Co-I
Keck/MOSFIRE	1.5 nights	MOSFIRE Survey of Galaxies in Early Rich Environments	Co-I
Keck/KCWI	4.5 nights	KCWI Survey of Ultradiffuse Galaxies in the Field	Co-I
Keck/DEIMOS	3.5 nights	DEIMOS Survey of the JWST CEERS Field	Co-I
Keck/DEIMOS	6.5 nights	Constraining the Physics of Satellite Quenching at $z < 1$	Co-I

Media Coverage

Seven Scholars Honored for Academic Excellence and Advocacy, *UC San Diego Today* April 2026
Black in Galaxy Astrophysics, *Nature Astronomy* June 2025
Devontae Baxter, Star Detective, UC Irvine Physical Sciences Communications Feb 2021
Some Granular Columns Weigh Too Much, American Physical Society News March 2020

Professional Memberships

American Astronomical Society 2020-Present
National Society of Black Physicists 2024-Present

Skills

Technical Skills: Python, LaTeX, Unix/Bash, Mathematica, Git, ADQL/SQL, Astrostatistics, Machine Learning
Soft Skills: Scientific communication, project management, student-centered teaching, mentoring across differences
Foreign Languages: Spanish (fluent), French (intermediate)

References

Reference 1	Reference 2	Reference 3	Reference 4
Prof. Michael Cooper	Prof. Alison Coil	Prof. Gregory Rudnick	Prof. Ethan Nadler
cooper@uci.edu	acoil@ucsd.edu	grudnick@ku.edu	enadler@ucsd.edu
Ph.D. Advisor	Postdoctoral Mentor	Research Collaborator	Research Collaborator

Publications

*Summary: **18** publications; **240+** citations; h-index=**9** (see ADS library or Google Scholar)*

First and Second Author Papers (8 papers, 80+ citations):

8. S. Um, **D. C. Baxter**, E. Nadler to be submitted., "Linking Orbital History to the Quenching of Isolated Dwarf Galaxies"
7. **D. C. Baxter** et al., 2026, "What Becomes of *JWST*/NIRCam-selected High-Redshift Massive Galaxies?" *ApJ*, submitted
6. F. J. Mercado, **D. C. Baxter**, M. K. Rodriguez Wimberly, et al., 2026, "The Quenched Fraction of Satellites Around Simulated Milky Way-mass Galaxies" *ApJ*, 1004, 217
5. **D. C. Baxter** et al., 2025b, "Quantifying the Impact of Observational Incompleteness on Identifying and Interpreting Galaxy Protocluster Populations with the TNG-Cluster Simulation" *ApJ*, 990 225
4. **D. C. Baxter** et al., 2025a, "The Importance of Gas Starvation in Driving Satellite Quenching in Galaxy Groups at $z \sim 0.8$" *ApJ*, 979, 41
3. **D. C. Baxter**, et al. 2023, "When the Well Runs Dry: Modelling Environmental Quenching of High-mass Satellites in Massive Clusters at $z \gtrsim 1$" *MNRAS*, 526, 3716
2. **D. C. Baxter**, et al. 2022, "The GOGREEN Survey: Constraining the Satellite Quenching Timescale in Massive Clusters at $z \gtrsim 1$" *MNRAS*, 515, 5479
1. **D. C. Baxter**, et al. 2021, "A Machine Learning Approach to Measuring the Quenched Fraction of Low-Mass Satellites Beyond the Local Group" *MNRAS*, 503, 1636

Co-authored Papers (10 papers; contributed to data acquisition/analysis and/or writing):

10. D. Sikorski, et al. (incl. **D. C. Baxter**), 2026, "The *HST*-Hyperion Survey: Environmental Imprints on the Stellar-Mass Function at $z \sim 2.5$" A&A, 708, A114
9. F. Giddings, et al. (incl. **D. C. Baxter**), 2026, "The *HST*-Hyperion Survey: Companion Fraction and Overdensity in a $z \sim 2.5$ Proto-supercluster" A&A, 706, A55
8. L. Sandoval Ascensio, et al. (incl. **D. C. Baxter**), 2025, "Caught in the Act of Quenching: A Population of Post-Starburst UDGs" *OJAp*, 8, 110
7. G. Hewitt, et al. (incl. **D. C. Baxter**), 2025, "Distinct Origins of Environmentally Quenched Galaxies in the Core and Virialized Regions of Massive Clusters at $0.8 < z < 1.5$" *MNRAS*, 541, 49
6. H. Gully, et al. (incl. **D. C. Baxter**), 2025, "Insights into Environmental Quenching at $z \sim 1$: An Enhancement of Faint, Low-mass Passive Galaxies in Clusters" *MNRAS*, 539, 3058
5. G. Gururajan, et al. (incl. **D. C. Baxter**), 2025, "Gas Properties as a Function of Environment in the Proto-supercluster Hyperion at $z \sim 2.45$" A&A, 698, A312
4. L. Xie, G, et al. (incl. **D. C. Baxter**), 2024, "The First Quenched Galaxies: When and How?" ApJL, 967, L42
3. E. Kukstas, et al. (incl. **D. C. Baxter**), 2023, "The GOGREEN Survey: A Critical Assessment of Environmental Trends in Cosmological Hydrodynamical Simulations at $z \approx 1$" MNRAS, 518, 4782
2. M. K. Rodriguez Wimberly, M. C. Cooper, **D. C. Baxter**, et al., 2022, "Sizing from the Smallest Scales: The Mass of the Milky Way" *MNRAS*, 513, 4968
1. S. Mahajan, M. Tennenbaum, S. Pathak, **D. C. Baxter**, et al., 2020, "Reverse Janssen Effect in Narrow Granular Columns" PRL, 124, 128002

Other Publications (1 white paper):

1. The Fellows of the National Osterbrock Leadership Program (incl. **D. C. Baxter**), 2023, "Reimagining the Astronomy PhD for the 21st Century" white paper, *American Astronomical Society*